

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

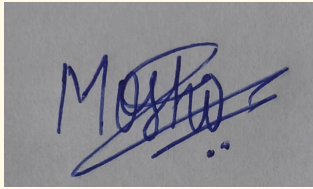
Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.Tech (AI-DS)	Date:	22.08.2026

Code of Conduct

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student:

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform Reference Resolution by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

ANSWER:

Aim

To design an algorithm that identifies pronouns in a discourse and links them to their correct antecedents using grammatical and contextual information.

Input

“Ravi met Arun at the library. He borrowed a book and later returned it.”

Algorithm

1. Read the complete discourse.
2. Divide the discourse into sentences.
3. Identify all noun phrases/entities.
4. Identify pronouns such as **he, she, it, they, them**, etc.
5. For every pronoun, create a list of possible antecedents from the previous context.
6. Check **number agreement**:
 - He/She → singular
 - They → plural
 - It → singular/non-human
7. Check **gender compatibility** where applicable.
8. Check grammatical position and recency.
9. Use semantic context to determine whether the candidate makes sense.
10. Select the candidate with the highest contextual compatibility.
11. Replace the pronoun with its corresponding entity.
12. Generate the resolved discourse.

Pseudocode

BEGIN

Input discourse

Split discourse into sentences

Identify all entities

Identify all pronouns

FOR each pronoun P:

Find possible antecedents before P

FOR each candidate C:

Check number agreement

Check gender agreement

Check grammatical role

Check recency

Check semantic compatibility

Select candidate with highest score

Replace P with selected candidate

Output resolved discourse

END

Application

The entities in the discourse are:

- **Ravi**
- **Arun**
- **book**

The pronouns are:

- **He**
- **it**

Resolution of “He”

Possible antecedents:

He → Ravi

He → Arun

Both Ravi and Arun are grammatically possible male entities. However, **Ravi is the subject of the previous sentence**, making him the more prominent discourse entity.

Therefore:

He → Ravi

Resolution of “it”

Possible antecedents include:

it → Ravi

it → Arun

it → book

The verb “**returned**” strongly indicates an object that can be returned. Since the immediately preceding object is **book**, the most semantically appropriate antecedent is:

it → book

Resolved Discourse

“Ravi met Arun at the library. Ravi borrowed a book and later returned the book.”

Discourse Representation

Ravi —————→ He

|

└─ borrowed —→ Book

↑

|

it

Validation

The resolution is correct because the algorithm considers more than simple word proximity.

For “**He**”, discourse prominence and grammatical role favor Ravi.

For “**it**”, semantic compatibility is especially important. A person is unlikely to be the object of “returned it” in this context, whereas a **book** is naturally something that can be borrowed and returned.

Conclusion

Reference resolution improves natural-language understanding by determining what pronouns refer to. It uses **grammatical agreement, recency, discourse prominence, and semantic compatibility** to establish correct links between pronouns and entities.

2. Develop an algorithm to analyze Text Coherence and Discourse Structure by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

ANSWER:

Aim

To identify logical relationships between sentences and construct a discourse structure showing how the sentences are connected.

Input

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Algorithm

1. Take the complete discourse as input.
2. Divide it into individual sentences.
3. Extract important events and concepts from every sentence.
4. Identify discourse connectives such as:
 - therefore
 - because
 - however
 - after
 - consequently
5. Compare consecutive sentences.
6. Determine whether the relationship is:
 - Cause-effect
 - Sequence
 - Elaboration
 - Contrast
 - Explanation
7. Create links between related sentences.
8. Construct a discourse tree or graph.
9. Verify whether the complete discourse forms a logical chain.

Pseudocode

BEGIN

Input discourse

Split discourse into sentences

FOR each sentence:

 Extract events and key concepts

 Identify discourse markers

FOR each pair of consecutive sentences:

 Analyze semantic relationship

 IF consequence marker exists:

 Relation = CAUSE-EFFECT

 ELSE IF temporal relation exists:

 Relation = SEQUENCE

 ELSE IF additional information is provided:

 Relation = ELABORATION

 ELSE:

 Relation = OTHER

Construct discourse structure

Output relations and structure

END

Sentence Analysis

Sentence 1

 “The roads were flooded after heavy rainfall.”

Main event:

Heavy rainfall → Roads flooded

This itself contains a **cause-effect relationship**.

Sentence 2

“Therefore, schools were closed for the day.”

The word “**Therefore**” is an explicit discourse marker.

Relationship:

Roads flooded → Schools closed

This is a **cause-effect/consequence relation**.

Sentence 3

“Students attended classes online.”

This describes what happened after the schools were closed.

Relationship:

Schools closed → Students attended online classes

This represents a **result/sequence relation**.

Discourse Structure

Heavy Rainfall



Roads Flooded



Schools Closed



Students Attend Online

Another representation:

S1: Roads were flooded



| Cause



S2: Schools were closed

|

| Result / Sequence

↓

S3: Students attended online classes

Relation Table

Sentence Pair	Relation	Explanation
S1 → S2	Cause-Effect	Flooded roads resulted in school closure
S2 → S3	Result/Sequence	School closure resulted in online attendance
S1 → S3	Indirect Cause-Effect	Flooding ultimately led to online classes

How Coherence is Maintained

The discourse is coherent because every sentence contributes to the same situation.

The progression is:

Rainfall → Flooding → School Closure → Online Classes

There are no unrelated statements. Each event provides a reason or consequence for the following event.

The word “**Therefore**” also explicitly signals the relationship between the first two sentences.

Conclusion

Discourse coherence allows readers to understand how individual sentences form a meaningful whole. Cause-effect and sequential relationships create a logical flow and make the text easier to understand.

3. Design an algorithm for a Conversational Agent that identifies Dialogue Acts from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

ANSWER:

Aim

To design a conversational-agent algorithm that identifies the communicative intention of each utterance.

Input Conversation

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Dialogue Acts

The main dialogue acts are:

- **Request** – asks the agent to perform something.
- **Question** – requests information.
- **Inform** – provides information.
- **Confirmation** – confirms completion.
- **Action** – actual task performed by the agent.

Algorithm

1. Receive an utterance.
2. Identify the speaker.
3. Tokenize and preprocess the utterance.
4. Identify keywords and sentence structure.
5. Determine the communicative intention.
6. Assign a dialogue-act label.
7. Maintain conversation history.
8. Use previous dialogue acts to interpret the current utterance.
9. Perform the required action if sufficient information is available.
10. Generate an appropriate response.

Pseudocode

BEGIN

Input utterance

Identify speaker

IF utterance asks agent to perform an activity:

 Act = REQUEST

ELSE IF utterance asks for information:

 Act = QUESTION

ELSE IF utterance provides information:

 Act = INFORM

ELSE IF utterance confirms completion:

 Act = CONFIRMATION

ELSE IF system performs requested task:

 Act = ACTION

Store dialogue act

Use conversation history

Generate appropriate response

END

Application

Utterance 1

 “Can you book a train ticket for me?”

Intent:

User wants the agent to book a ticket.

Dialogue Act:

REQUEST

Utterance 2

“Sure, where would you like to travel?”

The agent requires additional information.

Dialogue Act:

QUESTION

Utterance 3

“I want to go to Chennai.”

The user provides the requested destination.

Dialogue Act:

INFORM

Information:

Destination = Chennai

Utterance 4

“Your ticket has been booked.”

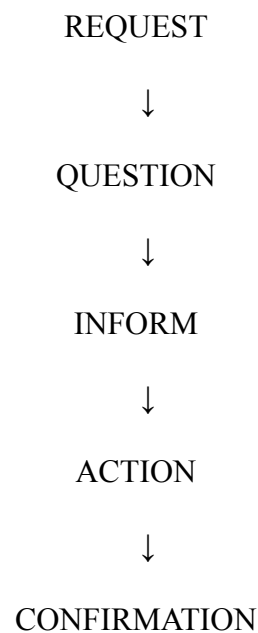
The agent indicates successful completion.

Dialogue Act:

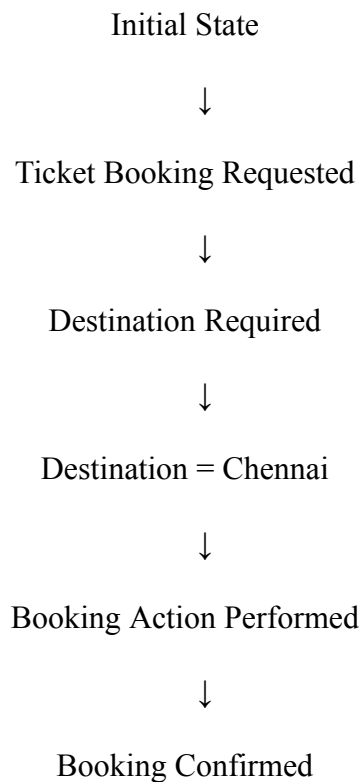
CONFIRMATION

The actual booking is the **Action** performed by the system.

Dialogue-Act Sequence



State Representation



Evaluation

Dialogue acts help the conversational agent understand **what the user wants rather than simply what the user says**.

For example:

"Can you book..." → Request

"Where..." → Question

"I want to go..." → Inform

"Booked." → Confirmation

The agent can therefore maintain a conversation state and ask for missing information before performing the requested action.

Conclusion

Dialogue-act recognition is an important component of conversational AI. It enables the system to distinguish between **requests, questions, information, actions, and confirmations**, allowing it to respond appropriately and maintain meaningful conversations.

4. Design an algorithm for Language Generation that converts a semantic representation into a grammatically correct sentence through Surface Realization. Consider the semantic input:

Action: Buy
Agent: Student
Object: Book
Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

ANSWER:

Aim

To convert a semantic representation into a grammatically correct natural-language sentence.

Semantic Input

Action = Buy

Agent = Student

Object = Book

Tense = Past

Algorithm

1. Read the semantic representation.
2. Identify the main predicate/action.
3. Identify the semantic roles:
 - Agent → Subject
 - Object → Object
4. Perform lexical selection.
5. Convert the action into the required tense.
6. Select appropriate determiners.
7. Arrange the words according to English syntax.
8. Perform morphological changes.
9. Generate the final surface sentence.
10. Check grammatical correctness.

Pseudocode

BEGIN

Input semantic representation

Identify Agent

Identify Action

Identify Object

Identify Tense

Select lexical words

IF Tense = Past:

 Convert Action into past form

Arrange:

 Subject + Verb + Object

Add appropriate articles/determiners

Perform surface realization

Check grammar

Output sentence

END

Step 1: Semantic Interpretation

Agent = Student

Action = Buy

Object = Book

Tense = Past

Therefore:

Subject = Student

Verb = Buy

Object = Book

Step 2: Lexical Selection

Semantic concepts are converted into words:

Student → "student"

Buy → "buy"

Book → "book"

Step 3: Tense Transformation

The input specifies **Past Tense**.

Therefore:

buy → bought

Step 4: Sentence Structuring

English commonly follows:

Subject + Verb + Object

Therefore:

Student + bought + book

Step 5: Determiner Selection

Appropriate determiners are added:

The + student + bought + a + book

Final Output

The student bought a book.

Surface Realization Structure

Semantic Representation



Lexical Selection



Morphological Processing



Syntactic Structuring



Determiner Insertion



Surface Realization



"The student bought a book."

Grammatical Validation

Component	Generated Text	Function
Determiner	The	Identifies noun
Subject	student	Agent
Verb	bought	Past-tense action
Determiner	a	Indefinite article
Object	book	Object of action

The generated sentence satisfies:

- Correct subject-verb structure
- Correct past tense
- Correct word order
- Correct use of articles
- Correct semantic meaning

Conclusion

Surface realization converts an abstract semantic representation into a grammatically correct sentence. It involves **lexical selection, syntactic structuring, morphological processing, article selection, and grammatical validation.**

5. Design an algorithm that combines Interlingua-based Machine Translation and Statistical Translation Approaches. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. **Analyze the source sentence.**
2. **Create an Interlingua representation.**
3. **Generate candidate target-language translations.**
4. **Apply statistical scoring to select the most appropriate translation.**
5. **Produce the final translated sentence.**

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

ANSWER:

Aim

To combine an **Interlingua-based approach** with **statistical translation** to produce an accurate target-language translation.

Source Sentence

"The boy is playing football."

For demonstration, the target language is **Tamil**.

Algorithm

1. Accept the source sentence.
2. Perform lexical and syntactic analysis.
3. Identify subject, action, object, tense, and aspect.
4. Convert the meaning into an Interlingua representation.
5. Generate possible target-language translations.
6. Calculate translation probabilities.
7. Calculate language-model probabilities.
8. Combine the scores.
9. Select the highest-scoring candidate.
10. Perform grammatical validation.
11. Output the final translation.

Pseudocode

BEGIN

Input source sentence

Perform lexical analysis

Perform syntactic analysis

Perform semantic analysis

Create Interlingua representation

Generate candidate translations

FOR each candidate:

 Calculate translation probability

 Calculate language model probability

 Calculate combined score

Select candidate with highest score

Perform grammatical validation

Output final translation

END

Step 1: Source Sentence Analysis

Sentence:

“The boy is playing football.”

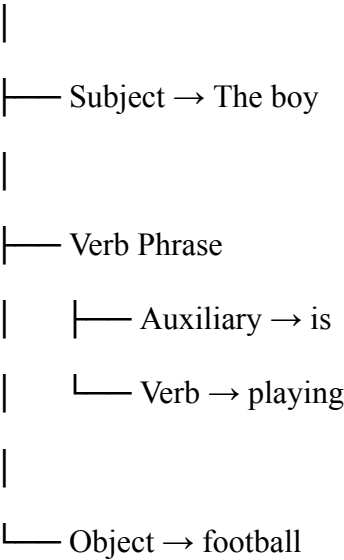
Lexical Analysis

Word	Role
The	Determiner
boy	Noun
is	Auxiliary
playing	Verb

football	Noun
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Syntactic Structure

Sentence



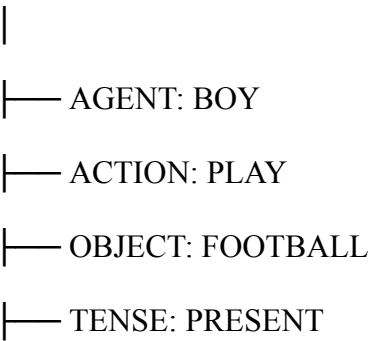
Semantic Analysis

- Agent = Boy
- Action = Play
- Object = Football
- Tense = Present
- Aspect = Continuous

Step 2: Interlingua Representation

The source sentence is converted into a language-independent representation:

EVENT



└─ ASPECT: CONTINUOUS

This representation focuses on **meaning rather than English word order**.

Step 3: Candidate Translation Generation

Possible Tamil candidates can be generated:

Candidate 1:

சிறுவன் கால்பந்து விளையாடுகிறான்.

Candidate 2:

சிறுவன் கால்பந்து விளையாடினான்.

Candidate 3:

சிறுவன் கால்பந்து விளையாடுவான்.

These candidates represent different temporal interpretations.

Step 4: Statistical Scoring

A statistical system estimates:

Translation Probability

+

Language Model Probability

A simplified scoring function can be represented as:

$$\text{Score}(T) = P(T|S) \times P(T)$$

where:

- **S** = source sentence
- **T** = candidate translation
- **$P(T|S)$** = probability of translating source S as T
- **$P(T)$** = probability that T is a natural sentence in the target language

The candidate that best preserves the source meaning while sounding natural in the target language receives the highest score.

Example

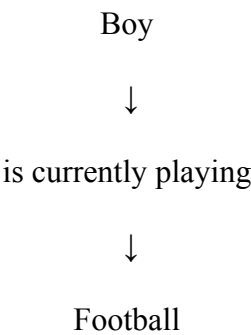
Candidate	Meaning Match	Fluency	Overall
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சிறுவன் கால்பந்து விளையாடுகிறான்	High	High	Highest
சிறுவன் கால்பந்து விளையாடினான்	Medium	High	Lower
சிறுவன் கால்பந்து விளையாடுவான்	Medium	High	Lower

Step 5: Final Translation

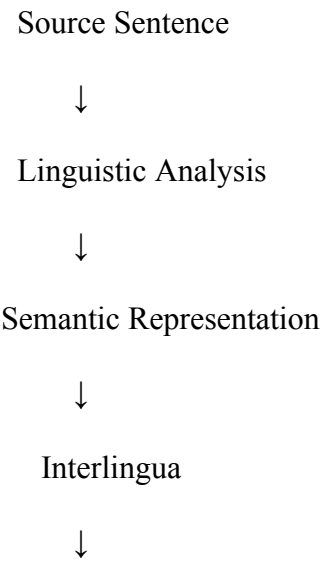
சிறுவன் கால்பந்து விளையாடுகிறான்.

Meaning



The present-continuous meaning of the English sentence is preserved.

Hybrid Translation Architecture



Candidate Generation



Statistical Scoring



Best Candidate Selection



Grammatical Validation



Target Translation

Advantages of Interlingua

1. Language Independence

The Interlingua representation does not depend heavily on English grammar.

English → Interlingua → Tamil

The same representation can potentially be used for multiple languages.

2. Reduced Structural Ambiguity

Instead of directly translating individual words, the system first identifies:

WHO → Boy

DOING WHAT → Playing

OBJECT → Football

WHEN → Present/Continuous

This helps preserve meaning.

3. Better Handling of Word Order

Different languages use different word orders. Interlingua separates semantic relationships from source-language word order.

4. Statistical Improvement

Statistical scoring helps choose the most natural translation when several candidates are possible.

5. Better Fluency

The language model favors target-language sentences that are grammatically and statistically more natural.

Evaluation

The hybrid approach combines the strengths of two techniques:

Technique	Main Contribution
Interlingua	Preserves semantic meaning
Candidate Generation	Produces possible translations
Statistical Model	Ranks possible translations
Language Model	Improves target-language fluency
Grammar Validation	Checks correctness

Final Conclusion

A hybrid **Interlingua + Statistical Machine Translation** system first converts the source sentence into a language-independent semantic representation. This reduces problems caused by differences in grammar and word order. Statistical scoring then evaluates possible translations and selects the candidate that provides the best combination of **meaning preservation, translation probability, and target-language fluency**. For the given sentence, the final Tamil translation is:

சிறுவன் கால்பந்து விளையாடுகிறான்.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____